

Igor Pereira Gouveia

Computer Engineering

📍 Fortaleza - CE | 📞 (85) 985201603 | ✉️ @igorpr03 | 🐱 IgorPrGv | 📧 igorprgy18@gmail.com

📄 CV Lattes | 🔗 LinkedIn | 📁 Portfolio

SUMMARY

Computer Engineering student at the Federal University of Ceará, with experience in software development and IoT systems. At GREat, I work on a health monitoring platform, contributing to the Java backend, React interfaces, and the integration of a Python-based ML service for biometric data analysis. I am seeking opportunities in backend or full-stack development, building data-oriented systems and integrating Machine Learning capabilities into real-world applications.

EXPERIENCE

Undergraduate Research Fellow (CNPq) | Backend Developer

GREat (Computer Networks, Software Engineering and Systems Group)

09/2025 - Present

- I work on the development of an **IoT platform for health monitoring**, which brings together clinical questionnaires and data collected from smartwatches, such as sleep and heart rate, into a monitoring flow for healthcare professionals.
- I work on the **Java backend** with **Spring Boot** and on **React** interfaces, contributing to authentication flows, patient-professional connections, questionnaire management, and data visualization.
- I participate in the development of an **ML pipeline** involving EDA, temporal feature engineering, model comparison, and integration through a **Python microservice** with **FastAPI** to generate quality-of-life and mental health indicators.

Undergraduate Research Fellow (Funcap) | IoT Developer

NuTec (Center for Industrial Technology and Quality of Ceará)

04/2025 - 08/2025

- I developed an **automated irrigation IoT prototype** with **ESP32** to reuse condensed water from air-conditioning systems, in a scenario with an estimated collection potential of 46 to 90 L/day.
- I implemented the control logic in **C++**, integrating soil moisture and reservoir level sensors, pump activation, MQTT telemetry, **Supabase** persistence, and a **Grafana** dashboard for real-time monitoring.

Embedded Systems Developer

LESC (Computer Systems Engineering Laboratory)

12/2024 - 01/2026

- I worked at **Clube do Hardware**, an extension project at UFC, developing educational prototypes in **C/C++** with **ESP32** or **STM32**, integrating sensors and modules into IoT projects. I also managed teams during the execution of two projects.

Web Developer (Frontend and Backend) | Marketing Manager

GTI Engenharia Jr.

12/2023 - 02/2025

- I worked as a web developer delivering more than four **institutional websites** and **on-demand systems** (e.g., blogs), using the **MERN stack** with MongoDB, Express, React, and Node.js to implement features and interfaces.
- As **Marketing Manager**, I led the team during **prospecting** and **lead qualification**. In addition, I structured commercial proposals, conducted more than five **negotiations**, and monitored alignment between scope and the technical execution.

EDUCATION

B.Sc. in Computer Engineering

03/2022 - 12/2026

Federal University of Ceará (UFC)

Vice-President, IEEE CASS Student Chapter (UFC)

09/2025 - Present

- I participate in a study and research initiation group focused on FPGA, with practice in HDL and Verilog and study of digital hardware design flows.

Teaching Assistant, "Applied Computational Intelligence"

09/2025 - 01/2026

- I supported around 20 students with theoretical and Python programming questions on topics such as exploratory data analysis, regression, classification, neural networks, and genetic algorithms, in addition to helping with the scientific writing of course reports.

SKILLS

- **Languages:** English (B2).
- **Programming Languages & Tools:** C/C++, Python, TypeScript, Verilog, Git, Linux, Docker, Kubernetes.
- **Data Science & ML:** Pandas, PyTorch, OpenCV, CrewAI.
- **Software & Databases:** React, Node.js, Express.js, MongoDB, PostgreSQL, Prisma ORM, Spring Boot, FastAPI, AWS Services (EC2, SQS, SNS, RDS, DynamoDB, S3, MiniIO).

CERTIFICATIONS & COURSES

Computer Vision Track: Image and video processing and interpretation, CNNs, generative models, and MLOps fundamentals.

Eldorado Institute (Nexvisual)

05/2026 - 06/2026

Quantum Tech Track: Quantum computing, variational circuits, and introduction to quantum ML.

CESAR School & Eldorado Institute (Brazil Quantum Camp)

02/2026 - 03/2026

Fundamentals of Deep Learning: PyTorch with CNNs, data augmentation, transfer learning, and introduction to NLP.

Nvidia DL Institute (in-person workshop)

11/2024

Introduction to Data Science: Python and Pandas, exploratory data analysis, data preparation, and basic modeling.

Atlântico Institute (Avanti Bootcamp 2023.2)

11/2023

COMPETITIONS

- [2024](#) and [2025](#) ICPC South America Brazil First Phase (Honorable Mention)